

A Research Agenda for Extending Agile Practices In Software Development and Additional Task Domains

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Abstract

This article is intended to serve as an introduction to this special issue on agile practices. In doing so, we briefly survey a number of key issues that are emerging in the application of agile practices to software development (SWD) and, similarly, examine recent work on extending knowledge about these practices to other task domains. We note that the extant literature on agile practices has been criticized for lacking a theoretical basis and comment on various ways that a theory orientation can enhance the accumulation of knowledge in this area. We also address issues that expand our current understanding of agile practices as they apply to non-SWD tasks. We present a framework for surfacing and discussing some of these emergent issues. We comment on the articles in this special issue and situate them within the research framework. We discuss some of the topics we think are likely to become influential as agile practices move outside the SWD domain and, finally, we present some summarizing observations.

Keywords

agile practices, software development, traditional practices, theory-oriented research

Introduction

We are more than halfway through the second decade following the publication of the *Agile Manifesto* (Beck et al., 2001). In terms of Gartner's hype cycle¹ (Fenn & LeHong, 2011), the use of agile practices² has had time to pass well beyond the peak of enthusiasm and the valley of disillusionment to become a relatively stable part of the portfolio of project management practices. The ABI/Inform database (March 2018) lists 761 peer-reviewed scholarly articles based on the search for "agile software development" (SWD) since 1998³. Even a quick sample of these articles shows that a great deal is known about agile practices. The use of the different agile practices is well documented for SWD in literature reviews (Chuang, Luor, & Lu, 2014) and bibliometric analyses (Dingsøyr, Nerur, Balijepally, & Moe, 2012; Lechler & Yang, 2017). As a project management standardization body, the Project Management Institute (PMI) recognizes this trend and has integrated agile approaches to project management in its latest edition of *A Guide to the Project Management Body of Knowledge (PMBOK® Guide)* – Sixth Edition (Project Management Institute, 2017a). PMI has also published the *Agile Practice Guide* (Project Management Institute, 2017b), a separate practice guide dedicated to agile.

We discuss SWD as a distinct domain knowing that it is difficult to define with precision. On the one hand, a project to add particular features to a software package like SAP would clearly be SWD, whereas a project to build a marketing

campaign for a new food product would not be SWD. Between the two, however, are projects that heavily use software tools in the process of providing business value. Many analytics projects exemplify this, where heavy database access is combined with statistical analysis to provide user decision support in a particular domain such as marketing. Our focus in this article contrasting SWD and non-SWD task domains follows that research to date regarding agile practices, which have mostly been about clear software development. Our concern is primarily for projects in which software development is minimal. We recognize that a sizable number of projects include a substantial information technology component. For these mixed projects, findings from studies based on purely software projects or those completely absent of information technology should be extrapolated for application to the individual project at hand.

In this article, we focus on extant research and emergent opportunities deriving from agile practices applied to SWD projects and the transfer of these practices to other task

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domains. At this time, the field continues adding depth and breadth to the body of knowledge regarding agile practices to SWD projects. For example, Baskerville, Pries-Heje, and Madsen (2011) trace changes from “pre-agile” practices incorporating the beginning of agile practices into SWD and the evolution of agile practices and their organizational integration in particular settings. We also see beginning efforts in practice, gradually being mirrored by research, testing whether such practices can be successfully transferred to organizational projects in task domains outside of SWD. We briefly survey the current state relative to these research areas; present a framework for formulating additional questions, particularly about the transfer from SWD to other task domains; comment on the articles appearing in this special issue; and discuss emergent areas for future research. We do this in a context of supporting theory and empirical testing based research as a way to expand knowledge in a verifiable and cumulative manner.

Agile Practices in SWD

The increasing interest in agile SWD practices is reflected by the steadily growing number of academic publications per year (Dingsøyr et al., 2012; Lechler & Yang, 2017). Many practitioners and scholars have come to believe that applying agile rules to software development processes has provided higher levels of productivity (Cardozo, Neto, Barza, França, & da Silva, 2010) and improved efficiency (Li, Moe, & Dybå, 2010; Lindvall et al., 2002) than traditional approaches in at least some cases. At this point, we can legitimately say that a body of evidence has accumulated supporting the notion that agile approaches to project management and development processes can be successfully applied in SWD projects (Dybå & Dingsøyr, 2008) and achieve software project success (Mohagheghi & Jørgensen, 2017; Serrador & Pinto, 2015). This conclusion reaches back to early indicators (Reifer, 2002), indicating that agile practices *can* (but do not necessarily *always*) provide measurable benefits on traditional project outcome measures such as duration and cost. On the other hand, in critically reflecting on the principles and values of agile practices, Rakitin (2001), Boehm (2002), and Levine (2005) show ambiguities in their potential interpretation and show how, in some cases, organizations can be challenged to assimilate and use agile practices. This research suggests that the application of agile practices can lead to successful outcomes, contingent on the way they are applied.

Much agile research has been concerned with describing and testing different agile software engineering models, including test-driven development, eXtreme programming, pair programming, Crystal, and Scrum (e.g., Lechler & Yang, 2017; Paige, Chivers, McDermid, & Stephenson, 2005). Some of these papers focus on presenting new techniques or those that extend known techniques while showing through “proof of concept” demonstrations that the technique can work. Other papers consider the relationship of actual practices among agile developers and fidelity to the enunciated principles of agile

development (Williams, 2012). Interestingly, while Williams found many areas where respondents observed commonality between the agile “philosophy” and practice, “commenters also said the principles did not cover planning, learning, and collaboration, and communication was not emphasized enough (Williams, 2012, p. 75).”

Another major direction of research compares agile and traditional practices. Such studies have sometimes provided conceptual comparisons of the attributes of each approach (e.g., Stoica, Mircea, & Ghilic-Micu, 2013). Agile research is still occupied with comparisons between traditional and agile practices (Bishop, Rowland, & Noteboom, 2018; Bonner, Kulangara, Nerur, & Teng, 2016) to demonstrate the merits of agile adoption. Empirically testing the value of agile versus traditional practices is difficult on a project by project basis, because in practice no two projects are quite alike. Experiments in which project tasks can be held constant, are difficult to construct with enough complexity and realism to generalize easily to the practice environment. Advantages of shifting from traditional to agile practices are, therefore, typically assessed by looking at average costs and benefits before and after an organization shifts commitment from one to the other or by assessing pilot projects. Such measures may be more or less rigorous given (1) a tendency to justify the cost of transitioning, (2) a reluctance to return to old practices if new ones prove to have only a marginal positive or negative net effect, and (3) variations in relevant measures (e.g., agile practices create flexibility and customer satisfaction, but at the price of maintainability).

We see a growing interest focused on the role and effectiveness of specific agile practices (e.g., timing of iterations, daily stand-up meeting, refactoring) (Tripp & Armstrong, 2018). We also see a growing concern for scalability. For example, what is the impact on skill profiles of an IT workforce moving from traditional to agile approaches?

One theme recently attracting attention relates to applying agile practices to large-scale projects (Dingsøyr, Fægri, & Itkonen, 2014; Hobbs & Petit, 2017a, 2017b; Lous et al., 2018; Petit & Besner, 2013). As project deliverables get larger (installing a new global enterprise system, for example), the applicability of agile techniques may not be assured or need to evolve by modifying existing and adding additional practices (e.g., Ebert, 2018; Goh, Pan, & Zuo, 2013). Ebert (2018) describes issues that arise with applying agile development techniques rolling out global IS products. A related issue more generally focuses on cultural issues related to global IS projects. Balasubramaniam, Cao, Kim, Mohan, and James (2017) take a high-level view of culture by examining implementation of agile SWD in three Asian countries. The authors find significant and numerous facilitators and inhibitors of the agile approach when shifting from the primarily Western context where it was developed to use in Asia.

Yet another research topic pertains to the organizational tendency to combine practices from both traditional and agile

sources. Baskerville et al. (2011) detail various mechanisms for integrating agile and traditional practices often by defining different roles for each. While agile projects may use burn-down tools for specific tasks, another layer of more traditional project management defines benchmarks and tracks progress over time. Tripp, Riemenschneider, and Thatcher (2016) identify six specific project management techniques associated with agile (and an equal number of software development techniques) and measure the degree to which each is used by surveying team members. Such hybrids have been examined normatively in terms of individuals proposing particular models (e.g., Gill, Henderson-Sellers, & Niazi, 2018). Questions remain, however, pertaining to how organizations in fact design hybrids, whether there are practices necessary for success across traditional and agile practices, whether some hybrids outperform others across task domains, and whether there are some hybrids that excel in particular task domains but not others. Given the uncountable variations among hybrid instances, it will be challenging to select specific versions for rigorous testing across the spectrum of possible designs.

Though much knowledge has been developed regarding the use of agile practices in the SWD domain, the body of literature as a whole is not without its critics. Methodological issues have been observed for empirical studies evaluating agile practices, which concluded that empirical results often do not meet the basic standards of validity and reliability (Abrahamsson, Conboy, & Wang, 2009; Dybå & Dingsøyr, 2008). Consequently, the contribution is “very low” and “very uncertain” regarding the contribution of existing agile studies evaluating agile practices as they might influence related to their adoption (Dybå & Dingsøyr, 2008, p. 851).

Transferring Agile Practices to Other Task Domains

The challenge in defining what exactly differentiates SWD from other task domains is reflected in a study by Batra (2017), which examined the use of agile practices for projects involving both data warehouses and business analytics. The study concludes by differentiating methods for data warehousing and analytics projects. The study notes that data warehousing typically pertains to large and relatively permanent structures for gathering, integrating and storing data, whereas analytics tend to be shorter term projects focused on retrieving and manipulating corporate data often for only one time use (thus removing issues like documentation, security, and robust user interfaces). These project types are distinguished not only by size and duration, but also by the directness of customer involvement and the permanence of the created artifacts.

That there is interest in transferring agile practices outside of the SWD domain is clear. Abrahamsson et al. (2009) indicate research directions for establishing agile practices in organizational functions including finance, contracting, legal, and human resources. Agile approaches can be adapted to develop new products or businesses (Blank, 2013; Ries, 2013) and

various other project contexts (Conforto, Salum, Amaral, da Silva, & de Almeida, 2014; Ktata & Lévesque, 2009). Ries (2013) and Blank (2013) translate agile practices into the domain of new product development. Their work, however, comes largely from an SWD context in the sense that the new products often include a significant software component. The applicability of agile in new product development may be a function of the degree to which such a software component is included. Conforto et al. (2014) show the influence of some agile tactics for project managers in companies other than those producing software. It is not completely clear if agile was used in these companies to support SWD functions supporting other products or services or if the projects pertained to other task domains. Ktata and Lévesque (2009) take a somewhat different approach in suggesting that the agile processes of software design add more interconnection with the business processes that the software is intended to support. This topic is also applicable where there have been calls for scaling agile practices to the organizational level to create an “agile organization” (Rigby Sutherland, & Noble, 2018).

The idea of agility *per se* outside of SWD is not new (Conboy, 2009). Notions of agile and lean manufacturing have been prevalent since the 1990s (Potdar, Routroy, & Behera, 2017). In short, the concept of agile manufacturing is to maximize customer experience while shortening delivery time rather than minimizing cost. Following the literature review reported by Potdar et al. (2017), little consensus appears to be present in the literature about just exactly how such agility is achieved, what stands in its way, how the needs of different industries may affect the suitability of agile practices as a whole or on a component by component basis, and by what mechanism the generally reported financial and operational benefits are realized. Interestingly, Scrum, a technique associated with agile SWD processes, was mentioned in a number of articles as a mechanism for enabling agile manufacturing. The exchange of knowledge both from lean manufacturing and agile approaches to project management may be a source of ongoing growth and development of expertise in both arenas.

An intriguing question emerging from consideration of transferring agile practices from SWD is: Can we link better agile techniques with specific characteristics of the project task? For example, much of SWD is intangible in the sense that code represents an instantiation of an algorithm and, although it runs on hardware, it is generally independent of the physical world. In contrast, construction is largely tangible: if the bricks aren’t delivered, the masons have nothing to build. It is straightforward to extrapolate that a class of problems pertaining to the demands of working in the physical world will arise for agile practices in the construction industry while they may not be of much concern for SWD.

To this point, we have framed issues looking at the application of agile relative to SWD and non-SWD tasks. We might also reverse the focus and use issues long addressed in traditional project management studies to probe how such problems are (or are not) resolved using agile practices. We can look for

insights by considering how agile practices in principle and in practice address project selection, estimation of time and cost, risk management, stakeholder analysis, communication structures, and governance. These are areas in which heuristics and procedures have become commonplace. Posing the question directly about how each of these processes is addressed using agile practices can make explicit whether the same procedures are effective in both environments, whether the need for them disappears, and how they may be handled for SWD but not necessarily for other tasks. Ultimately, the question is: What is the fundamental nature of these problems universally across all task types?

Applying Theory

Much of the literature pertaining to agile SWD describes methods or particular instances of application of agile techniques, but does not necessarily provide theoretical generalization or explanation that would allow for projection or prediction in future cases (Conboy, 2009; Dingsøyr et al., 2012). Several comprehensive systematic reviews analyzed the growing body of research on agile practices and coincidentally bemoan this lack of theoretical understanding (Abrahamson et al., 2009; Goh et al., 2013; Hoda, Salleh, Grundy, & Tee, 2017). Movement toward a theory-oriented approach would mark a transition from a more design and engineering approach to a more behavioral science orientation. Dingsøyr et al. (2012) show a very small percentage of published papers using any theory base and, those that do, use a wide divergence of behavioral and mathematical theories. Few researchers have attempted to use theory building based on direct observation or inductive techniques, such as grounded theory methodology.

We propose three access points for using a theory-oriented approach to study the transfer of agile practices from SWD to other project domains:

1. Replicating in new domains the original atheoretical questions posed regarding SWD. This would entail examining (a) designs and techniques for applying agile practices in other domains. For example, such a design might show how the principle of work in relatively short iterations would be accomplished in construction or marketing projects; and (b) the larger effects of moving the organization's project orientation from traditional to agile. For example, what are the effects on strategic goals, agility of the organization, and portfolio-level effectiveness and efficiency? An accumulation of empirical data may suggest theoretical statements for further testing, thus creating a theory-building program.
2. Formulating statements based on SWD experiences or findings to test in the new domain. To the extent that particular practices of agile have created observable effects, we can hypothesize that these will also create the same effects in other domains. Implicit theory of an observed positive relationship between the use of specific approaches to agile features in project management features and project outcomes may be formally stated and tested. For example, if team member self-selection of tasks leads to more efficient operations in SWD, we can test whether it will lead to the same results for other task domains. In a sense, to the extent that such relationships are stated as theory, such a procedure could be viewed as testing the boundary within which it can be viewed as applying (Weber, 2012).
3. Formulating statements based on extant knowledge concerning project management techniques used broadly outside of SWD. For example, statements about the expected effects of using traditional risk management techniques being influential on project success may lead to hypotheses that agile practices will also address risk to the same extent or if agile practices address risk (even if in a different way) similar benefits will accrue. Alternatively, more standard management, psychology, and other discipline theories may potentially generate useful and testable hypotheses.

A Research Framework on Extending Agile Practices

It has been our intention to position this special issue at the intersection of the SWD and project management research fields. The original goals were to: (1) increase the breadth and depth of understanding the application of agile practices in SWD (beyond showing that it can be successful, what steps are or should be involved, and how it compares abstractly to comparison with traditional practices); and (2) to consider the state of research pertaining to the transfer of agile project practices to tasks outside of SWD. In this section, we reference extant literature and use it to pose emergent questions.

We propose a framework for surfacing new questions, particularly in regard to the transfer of agile practices outside the SWD domain. This categorization is aimed at surfacing questions and identifying new opportunities. No claim is made that the framework is comprehensive, that the categories are mutually exclusive, that all research study instances can be fit unambiguously into a single category, or that frameworks based on different dimensions might not also be useful for the same or other purposes.

This framework is based on two dimensions: level of analysis and functional perspectives of agile practice issues (Table 1). The level of analysis acknowledges a hierarchical view of issues when agile practices are instantiated in practice. We differentiate three levels of analysis: individual, team/project, and organization, following Staw, Sandelands, and Dutton (1981). The individual level is characterized by factors pertaining to individual team members, including skills and motivation. The team/project level refers to factors primarily addressing how groups of individuals are organized for collaboration and integration of work. Although the team

Table 1. Conceptual Review Framework

Level of Analysis/Application	Functional Perspectives			
	Behavioral Perspective	Process Perspective	Governance Perspective	Outcome Perspective
Organizational level Multiteam/multi-project level (program/portfolio)	Organizational behaviors	Agile in large projects Distributed agile Managing multiple approaches	Resource allocation and project selection Project management office	Aligning to organizational strategy Efficiencies
Single-team/single-project	Conflict management Team composition	Distributed work systems Stakeholder management	Role definitions Team autonomy	Iteration metrics Product contribution
Individual	Individual motivation Individual skills	Individual procedures and task fulfillment	Roles of actors Distribution of tasks	Performance evaluation Job satisfaction, retention, burnout

and project are treated together due to their frequent overlap, we recognize that there may be subtle differences between them. One team may work on multiple projects; one project may include work from multiple teams. We expect, however, that many issues pertain to both the team and project and treat them for purposes of surfacing research questions as a single category. The organizational level both initiates many (if not most) project tasks, typically provides resources, and is the environment in which most of the project products will be used. Conceptually, while it is helpful to look at these levels as distinct, there is clearly interaction among them, because individuals comprise teams and organizations and organizational context influence the nature of teams and the work of individuals. Indeed individual multilevel studies may explicitly measure variables at multiple levels and show their relationships.

The second dimension is based on four broad functional perspectives: behavioral, process, governance, and outcome. The behavioral perspective focuses on activities, actions, and attitudes. The process perspective targets abstract benchmarks, plans, and procedures used to guide behaviors. The governance perspective is comprised of control, coordination, reporting, and feedback mechanisms that organize and direct project activities. Agile practices tend to shift formal governance mechanisms to roles and responsibilities. Finally, the outcome perspective represents results of project implementation at all levels and affecting all stakeholders.

Together these two dimensions generate 12 distinct categories, each of which suggests a set of issues including those identified in Table 1. We discuss these issues by level of analysis.

Individual Level Analysis

Behavioral perspective. Individual-level behavioral issues include concerns regarding skills and motivations. Issues regarding skills for individuals working within agile practices remain relatively unexplored. Important research questions

from this perspective include: What are the minimum thresholds for task-related skills (e.g., in construction should the plumber need to be able to design systems or is it sufficient to install pre-organized ones?) or for team interactions (e.g., being able to accept non-optimal assignments if it increases team performance)? On the whole, one can expect those in agile environments to need a broader range of skills with less depth for each since each team is handling, to some extent on its own, a wider range of tasks. In the SWD environment this may mean that database specialists end up testing or debugging code. But does it mean that the carpenter also helps install pipes?

We also see motivation being an important element in individual and team performance. Will agile teams function effectively if not all members buy into the agile philosophy? Is there a threshold number of self-motivation or preference for agile needed among team members for effectiveness? Are there effective methods to establish a culture supporting agile where team members have commitment to traditional approaches? Do answers to these questions differ systematically based on task domain?

Scholars might also consider how agile practices hold up as generations of new workers (e.g., baby boomers versus millennials) exhibit different work attitudes and preferences. In the longer run, shifts in technology and social structures may change how work is done (e.g., contracting with individuals, open source, free agents, crowdsourced) in ways that affect the organization of agile project teams and work arrangements within and outside the SWD domain.

Process perspective. In SWD agile practices, there are many flavors of specified processes. Do these work equally well relative to different skills, styles of work, and other individual characteristics? Do lessons along these lines hold across task domains? Some agile SWD practices aim to improve productivity by moving individual to group work (i.e., paired programming), but do these techniques work across task domains?

Governance perspective. In some sense, the essence of agile practices is shifting control of work from rules and checklists to roles and responsibilities, which tend to be relatively fixed, unambiguous, and stable in the traditional practice environment, but more flexible and varied in the agile environment. Although some team members will embrace this change, others may prefer to apply a narrower and more predictable skill set. The role for scrum masters and other leaders in the agile environment may also be complex and require the development of expanded skill sets. Business representatives on agile teams may experience difficulty in prioritizing particular requirements and signing off on deliverables as they crop up. Are there new standard roles (perhaps incorporating or modifying some or all of the SWD roles) that would apply outside of SDW?

Outcome perspective. One set of outcomes on the individual level revolve around performance evaluation. Positive judgments about individual work should lead to higher quality future assignments, better compensation, and continued progression on a career ladder. However, such evaluation is difficult for all but the most structured tasks. This is particularly difficult where work is done in a strongly team-oriented environment in which the link between individual effort and performance is highly influenced by the actions of fellow team members. The essence of the problem can be phrased in terms of assessing the contribution individuals make to the product and process of its creation. We are not aware of research in the SWD domain that addresses this outcome issue. In considering the shift from SWD to other domains, can methods for evaluating the contributions of individuals be applied? Are there methods for evaluating individual performance in strongly group-oriented environments that can be transferred to agile development? Is there any need or utility in individual-level performance evaluation in any circumstance?

Other individual outcomes have provided a basis for investigating agile practice effects. Research in the SWD realm suggests that using agile practices leads to more satisfied employees (Highsmith & Highsmith, 2002). Another study found that, compared with plan-driven models, software developers working in agile contexts experienced higher job satisfaction (Melnik & Maurer, 2006). Similar findings were produced by Kropp, Meier, Anslow, and Biddle (2018). Job satisfaction was shown to increase with the use of specific agile practices such as Pair Programming (Pedrycz, Russo, & Succi, 2011) or eXtreme Programming (Mannaro, Melis, & Marchesi, 2004). While provocative, such studies do not thoroughly address the mechanics by which this relationship is created. Because we have little understanding of the mechanisms by which this effect occurs, assuming it is indeed robust, we cannot rule out that (1) is it a product of the Hawthorne effect—the effect of change per se rather than the methods; (2) that it is experienced by the subset of workers likely to participate in pilot studies and motivated ideologically to shift to a different

method; or (3) that agile practices are designed and modified to accommodate the skill set of team members already in place rather than change the team members or their skills to accommodate the idea agile practices configuration.

One way to address this would be to study such effects using the large body of management research applied to workers in general. Two of the most well-known job satisfaction models are Herzberg's two-factor model (also called Motivation-Hygiene theory) (Herzberg, Mausner, & Snyderman, 1959), and Hackman and Oldham's Job Characteristics Model (JCM) (Hackman & Oldham, 1980). Many relevant aspects of a job can be organized into intrinsic and extrinsic motivational rewards (Ryan & Deci, 2000). Intrinsic rewards refer to the inherent features of work and characteristics associated with the task; they tend to be less tangible, but more emotionally based for example, the sense of achievement, recognition, responsibility, autonomy, or accomplishment (Kalleberg, 1977). The extrinsic rewards often refer to characteristics that are external to the tasks themselves and tend to be more tangible, such as pay, job security, resource adequacy, and benefits (Herzberg et al., 1959; Kalleberg, 1977). Based on the two-factor model, Hackman and Oldham introduced the Job Characteristics Model (JCM), which describes how job satisfaction is influenced by five job characteristics: autonomy, task identity, task significance, skill variety, and feedback (Hackman & Oldham, 1980). Either model might explain *why* software developers experience higher satisfaction levels when working in agile environments.

Team-Level Analysis

A broad study by Conboy, Coyle, Wang, and Pikkarainen (2011) emphasized a variety of issues revolving around the team and project levels. The study examined 17 case studies of organizations using agile development practices over at least three years each. The study illuminated a broad array of issues pertaining to the management of teams. They conclude that "agile methods' increasing prevalence, the lowering of traditional agile boundaries, and growing pressure to adopt agile methods all contribute to the need for human resources departments and project managers to address associated skill and people challenges (p. 48)." More specifically they found that individual team members (1) needed a broader skill set and were less able to focus on a particular specialty; (2) experienced added pressure regarding communication—team members, for example, at times would reveal trade secrets and other protected information to clients; (3) developed knowledge of the business processes being supported through the project activities; (4) varied in individual commitment and acceptance of the "agile philosophy"; (5) can be poor at self-organizing decision making—may choose tasks for which they were not well suited; (6) continued to use individual performance evaluation, which is misaligned with team-oriented activities; and (7) faced difficulties in recruiting employees to project teams.

Examples of efforts to address these issues are presented throughout discussions of the cases. Methodical testing of various interventions (singly and in combination) and their effects on these various issues would likely be a productive direction for future research. Note that, as the authors point out, these problems don't tend to crop up when agile is a small segment of a larger organizational portfolio and team members self-select for inclusion but may become larger issues when the practice is scaled to organizational strength. New problems can be the fruit of initial success.

Behavioral perspective. The behavioral perspective at the team level focuses on the descriptive observation of what teams actually do, rather than the normative aspect of what they should do or benchmarks of a general abstraction of the sorts of things they do. Issues include conflict management and planning, which still occur in agile practices, even if differently from those in traditional project management settings. Assuming that both conflict and conflict resolution differ when shifting from traditional to agile practices, do those differences continue to move from SWD to other project types? A traditional project manager may minimize conflict through thoughtful interaction with individual members; where teams move toward the self-governing, the decrease in such conflict resolution may result in issues that affect productivity. Similarly, in the case of planning, the shift from traditional to agile likely will eliminate much of the singular long-term planning characteristic of the traditional, but this can be expected on occasion to lead to pursuit of limited value alternatives longer than necessary, hence the importance of refactoring in SWD.

Process perspective. Agile models emphasize collaboration between self-managing cross-functional teams (Conboy & Fitzgerald, 2004; Fowler & Highsmith, 2001). Several team-based models, such as XP and Scrum were developed to organize teams for developing software. These models cover different team processes including sequences of execution processes, work assignments, coordination, and communication. More specifically, it is held that these models move team members to minimize planning steps, improvise execution steps (when necessary), and accelerate the creation of interim products that clients can use even as larger systems are being developed. How do these models translate from SWD to other task domains?

Considering the project as a unit of analysis, moving to agile practices elevates the role of the customer and user participation in the development process. Are there models and processes for integration of varied participants within project teams (e.g., resolve conflicts between stability and urgency) and do these models translate from SWD to other task domains? Formal stakeholder analysis and management techniques can provide a basis for addressing varied purposes and goals relative to project deliverables. Are there tools and procedures for balancing the targeted and achieved outcomes across a range of stakeholders? Similarly, with risk

management, we see traditional project management approaches to the recognition and remediation of risk through the use of forecasting and estimation of loss and probability of occurrence. Does the need for formal risk management disappear in the SWD environment, but return in other task domains?

Governance perspective. Agile practices specify to a large degree specific team member roles. The scrum master, for example, defines a set of actions and responsibilities for those who take on this role. Of course the role becomes more complex when teams focused on the same project or deliverable are not collocated (e.g., Srivastava & Jain, 2017). The role of the customer is also somewhat defined and it is expected that customers are integrated into the development process to give immediate feedback. How does this differ from traditional project management across task domains? Efforts to translate the roles defined by specific agile practice components need to consider how traditional skill sets map into these roles, how individuals can develop their skill sets to meet these requirements, and the degree to which these differ across task areas. The way in which teams are formed may also be influential in generating project outcomes. Lee and Xia (2010) examine the effect of autonomy and the diversity of groups' ability to generate responsiveness and ultimately traditional project outcomes such as cost and schedule. This article exemplifies the opportunity of study in the SWD domain to set a baseline from which to test the applicability of agile practices in other task areas.

Outcome perspective. Performance measures at the team and project levels are likely to continue emphasizing the "iron triangle" of cost, schedule, and quality/scope. With an agile practice perspective, however, additional measures might include the degree to which teams and projects achieve agility-related goals, for example, the rate at which alternative designs are considered or the speed with which errors are discovered and remedied.

Some studies have examined team issues across functional issues. Maruping, Venkatesh, and Agarwal (2009) and Maruping, Zhang, and Venkatesh (2009) examined the relationship between collective ownership, coding standards, expertise coordination, and technical quality outcome. They found that collective ownership and coding standards affected technical quality and moderated the relationship between coordination and quality. Thus their investigation, while team-oriented overall, considered governance (coordination and ownership), process (standards), and outcome (technical quality) all in the context of the particular XP process. Strode (2016) focused on knowledge, process, and resource dependencies among agile team members to discover numerous linkages where coordination can present difficulties.

One promising theoretical basis for the study of team behaviors in general pertains to "information richness," also known as "media richness theory." Originally put forward by Daft and Lengel (1986), this theory proposes that the most efficient

group communication occurs when there is a fit between the richness “supplied” by communication media and task “demand” for such richness. Examination of group effectiveness, particularly those spanning geographic locations, can be forwarded by consideration of their intended and actualized patterns of communication. The general principles of this line of study have been applied to distributed groups and the use of media tools for coordination and collaboration among non-collocated team members (e.g., Prasad, DeRosa, & Beyerlein, 2017; Smit, Bond-Barnard, Steyn, & Fabris-Rotelli, 2017; Espinosa, Cummings, & Pickering, 2012).

Organizational-Level Analysis

The organizational level is the least developed level of analysis for research to date on agile practices. It stands to reason that the bulk of attention pertaining to agile approaches to project management has focused on the team and project and the work carried out by individuals within these. Issues on the organizational level largely pertain to the environment in which agile projects can be nurtured or inhibited.

Behavioral perspective. The behavioral perspective at the organizational level focuses on what organizations actually do regarding human resource policies pertaining to workforce management. This involves a variety of human resource management issues including: executing the transition between traditional and agile approaches, job redesign, performance evaluations, career paths, training and development, and grievance procedures.

Process perspective. Organizations, whether purposefully or by default, create and maintain structures regarding the methods of communication, decision making, and grouping of staff into work units. Standard and well-known tools for describing such structures include organizational charts and flow diagrams. Questions in this regard pertain to whether the instantiation of agile practices either have an effect on choices for how these structures are organized or, conversely, whether agile practices thrive differentially based on different sorts of organizational structures. More specifically, do firms benefit from creating buffer zones between traditional and agile practices? Do firms gain a net benefit from having a single approach, whether or not it is optimal for all projects, rather than incur the costs of maintaining multiple approaches?

Governance perspective. Significant issues pertain to selecting and allocating resources for projects in the agile environment. Traditional organizational managers will have much experience with formal models for proposing and evaluating project proposals based on return on investment and other financial models. It is unlikely that a set of equivalent models for such estimation are available in the agile environment; in fact, it is not clear that the need for formal estimation does not diminish agility. Cao, Mohan, Ramesh, and Sarkar (2013) present a hybrid model of buffering traditional and agile practices as

a way to address this issue. One of the advantages of agile is a flexibility with regard to tasks, which may translate into fuzziness with regard to estimation of project costs. Along similar lines issues pertain to the scaling of agile project practices to organizational integration of many projects in portfolios and programs, using, for example, scaling frameworks such as SAFe (Leffingwell, 2017). It is not clear how the flexibility of teams working on sets of task-related work packages can provide benefits from resource sharing across projects or how such work can be best coordinated while also being in the purview of individual team members. Given that project management offices (PMOs) have different charters and levels of authority, it will be of interest to better understanding how the variations in PMO are likely to influence the administration of agile approach projects.

Outcome perspective. One important lesson in SWD pertains to the difference between immediate or project costs and total cost of ownership. In some cases, saving money and time during project execution can result in weaker systems that require much more expensive lifetime maintenance. Costs that accrue outside of individual projects, such as team member training and development or supporting a PMO, may not show up as a simple sum of costs and benefits of individual projects. In moving from traditional to agile environments, the items comprising these additional costs may shift (perhaps be increased or reduced) and inadvertently become invisible for a period of time during the transition. Given the potential expense of changing an organizational traditional to agile environment, there may be significant potential for a “success bias,” meaning that anything but a clear-cut disaster will lead to continuation of the new program. It may also lead to biased analysis of the cost–benefit ratio, or even not measuring outcomes at all since the expenditures become sunk and there may be no interest in going back.

As with changes in cost items in transitioning from traditional to agile environments, intangible benefits may also accrue. For example, the alignment of project outcomes and organizational strategy may result in a set of deliverables at the same cost and schedule, but with varying levels of support for overall mission and goals. Even if the aggregate tangible costs of moving from traditional to agile environments are substantial, the ability to respond more quickly to shifting market conditions may render these dynamic capabilities a valuable investment rather than loss.

Special Issue Contributions

Although the original intent of the special issue was to focus on questions pertaining to the transfer of knowledge from SWD to other domains, we were also open to work that moved the understanding of agile practices forward within the SWD domain. In the end, most of the articles submitted remained firmly in the SWD domain. We consider each of the articles selected for the special issue in terms of the original question addressed from

Table 2. Matching Articles and Original Questions in the Call for Papers

Original Question from the Call for Papers	Article Authors and Title	Framework Positioning
From an organizational perspective, what are the trade-offs involved in shifting all project management to an agile approach, versus maintaining a mixed portfolio of agile and traditional development?	Sweetman and Conboy: Portfolios of Agile Projects: A Complex Adaptive Systems' Agent Perspective	Governance perspective at the organizational level
Are there project management practices that remain constant across traditional, hybrid, and agile approaches (e.g., risk management, stakeholder management, team building)? If not, how do best practices compare across approaches?	Lappi, Karvonen, Lwakatare, Aaltonen, and Kuvaja: Toward an Improved Understanding of Agile Project Governance: A Systematic Literature Review	Governance perspective at the team level
How are agile principles being applied in large projects and multisite projects?	Dingsøyr, Moe, and Seim: Coordinating Knowledge Work in Multiteam Programs: Findings from a Large-Scale Agile Development Program	Process perspective at the team level
Are there metrics and standards that can be used for control of agile project progress during execution? Can these be adapted from traditional project management or is there a need for creation of new metrics and standards?	van Oorschot, Sengupta, and Van Wassenhove: Under Pressure: The Effects of Iteration Lengths on Agile Software Development Performance	Process perspective at the team level
Are there process theory explanations for differentiating better from less successful ways to implement agile techniques?	Dennehy and Conboy: Identifying Challenges and a Research Agenda for Flow in Software Project Management	Process perspective at the organizational level

the call for papers (see Table 2); how it fits generally into the framework discussed above; and, more generally, how it addresses ongoing issues in the agile practices domain.

Sweetman and Conboy (in this issue) address the movement from individual group/project to their management as part of a larger program or portfolio. This article addresses governance perspectives particularly at the organizational level. The article also embraces a theory-oriented approach as it positions its study using complex adaptive structures (CAS) and, more broadly, complexity theory. More specifically, the authors address how adaptiveness can be achieved in the project portfolio. A set of 16 propositions are derived from 30 expert interviews aimed at the effective management of agile project portfolios. This work shows how by examining a broad range of cases, patterns can be detected that are not necessarily visible in each case.

Lappi, Karvonen, Lwakatare, Aaltonen, and Kuvaja (in this issue) provide a comparison of traditional and agile practices specifically in terms of governance practices. This article addresses governance perspectives primarily at the team/project level of analysis. Their comprehensive literature review tackles the range of governance practices where projects are managed with agile practices. They explore whether specific governance practices are similar between traditional and agile approaches to project management. By distinguishing structures found in agile, traditional, both, and neither, this article presents an implicit theoretical basis for examining effects on project outcomes at the practice level. It also presents a basis for further consideration of how to construct such structures, specifically for either approach in order to enhance either.

Dingsøyr, Moe, and Seim (in this issue) consider the scaling of agile practices to deal with large projects. This article focuses on a process perspective within the project/team level of analysis and examines the coordination aspect of large Scrum-based projects. Their study blends the specifics of work using a generally agile approach with the theoretical constructs associated with coordination. This study applied theoretical constructs drawn from the coordination arena to the particular large project domain; it also observed activities enriching our understanding of the nature of coordination per se as well as how it plays out in the use of agile approaches. More specifically, this study explores how multiple developer teams in large-scale SWD programs are coordinated. With an in-depth case study, they analyze how knowledge work is coordinated in a large-scale agile development program involving 12 teams. Their findings highlight how practice overcomes the complexities of coordinating multiple developer teams in an agile-based project implementation of large-scale projects by using multiple coordination mechanisms. The case also shows how the use of different coordination mechanisms changed over the different implementation phases, suggesting some contingencies for different coordination mechanisms in the agile arena.

van Oorschot, Sengupta, and Van Wassenhove (in this issue) address issues of metrics and standards, focusing largely on deadlines and iterations as a particular agile practice. This study takes a process perspective at the team/project level of analysis. This article primarily addresses the optimal length of iteration cycles in agile projects. Scrum is presented in the literature as a series of iterations. Each iteration cycle is limited by a date by which specific results have to be delivered. The

length of these iteration cycles is normally set to 20 working days, which is the normative assumption in the literature; however, its rationale is not clear. In their article, the authors question this normative assumption for iteration cycles and analyze with a complex simulation how the cycle length is affected under the influence of the different behavioral and procedural aspects of SWD. Based on their simulation, the authors suggest a much longer cycle time that should be closer to 43 to 65 working days. These results are also potentially relevant for traditional project management for the setting of milestones. Following this logic, project milestones should be at least two or three months apart. This article also has theoretical implications suggesting a [testable] u-shaped relationship describing the tensions between schedule pressure and project performance. It also suggests how that relationship is likely to change when the single deadline constraint is loosened. Another theoretical contribution is in suggesting that the amount of flexibility obtained must be finite.

Dennehy and Conboy (in this issue) present in a broad sense an alternative to both the traditional and agile practices. Although spanning across categories in our framework, we see this work primarily as a process perspective article. This is due to how it focuses on the structures and procedures of projects. Though it has aspects of defining particular projects, it is mostly about an organizational level approach to organizing projects more generally as practiced across multiple projects over a period of time. The article examines the production of software as a sequence of repeatable, operations management style activities applied to a set of projects moved along a structured workflow. In addition to describing what may be at the beginning stage of an emerging trend, the authors identify potential difficulties and challenges that may be expected to inhibit its diffusion. Such identification of difficulties presents an implicit theory, suggesting that if the approach does achieve wide distribution, we may look to see if indeed these potential barriers were “solved.”

Considering the collection of articles as a whole, we find that they add to our knowledge of agile practices, particularly relative to increased understanding as applied in the SWD domain. We see, however, undiminished opportunities for extending knowledge regarding the transfer of the agile approach to other task domains.

Emergent Topics

In this section we consider a number of additional topics potentially critical to the transfer of agile practices outside of SWD.

Integrating agile practices with physical materials. One of the obvious and main distinctions between SWD and other task domains pertains to the usage of physical materials. Given that domains such as SWD and construction are significantly different in their reliance on physical materials, does this substantially change the range of applications for agile practices? Secondly, as physical materials are utilized the planning and

timing of their application become critical. Can agile practices in task areas highly organized around physical materials account well enough for issues of coordination and timing? Although in SWD there is significant outsourcing and contracting for software components, the intangibility of the products may leave much room for variance in how they are created and what their internal structure might look like. Batra (2009) outlines what some of these potential problems might be and calls for empirical investigation. Might practices for SWD contracting be over engineered to the needs of contracting arrangements in more traditional task areas?

Diffusion patterns. Agile practices have spread widely and quickly in SWD practice; however, this does not seem to be the case in other task domains where the rate of diffusion seems much slower. Interest in this area is exemplified in the work of Cram and Newell (2016) and Wang, Conboy, and Pikkarainen (2012). Both studies examine how agile practices are adopted by organizations. Cram and Newell report three levels of commitment by organizations to the adoption of agile practices, ranging from a surface level that is not destined to last long to a thorough integration within the firm. This line of inquiry is of interest (1) in historical terms to examine how the practices have spread—in terms of communication patterns, messages communicated, and evaluation of trials; (2) in terms of thresholds of absorption and specific organizational precursors to adoption such that the agile practices become stable and persistent; and (3) the relationship between the penetration of adoption and the realization of net benefits from agile practices. Are there organizational characteristics affecting their likelihood of adopting or being successful if they do? Wang et al. (2012) show a diverse set of non-linear patterns by which agile practices are institutionalized. Tripp and Armstrong (2018) examined precursors, such as project leader experience, in affecting the rate of adoption of agile practices. While there has been much commitment to the agile approach in SWD, Cram and Newell (2016) suggest the possibility that these practices represent a sort of management “fad,” which might have significant potential for practitioners to name their methods “agile” while straying far from the original agile conceptualization as found in the *Agile Manifesto* (Beck et al., 2001).

Hybrids outside of SWD. Can organizations adopting agile practices outside of SWD leapfrog trial and error with varied combinations of agile and traditional practices and go straight to a particular hybrid with elements that give it a high probability of being a good match for its intended use? Tripp et al. (2016) provide distinct categorizations of specific agile practices and software development techniques. By asking survey respondents about the amount to which each practice is used, the authors obliquely derive a potential loose taxonomy of hybrids. They also, by identifying both project management and software development techniques, provide a basis for examining

agility in project management outside of SWD simply by focusing exclusively on the development procedures.

Agile projects and organizational agility. The idea of agility has long been considered in the organizational context. It is widely held that the importance of organizations to react quickly to changes in market condition increases with environmental turbulence. Tallon (2011) links organizational agility as a desirable outcome in itself to operational decisions and configurations of software at a departmental level. Tallon shows that differentiated investments in agility across departments may result in more agility for the entire organization. Following this line of thinking, it is an open question as to whether agile project practices, inside and beyond SWD, result in more agile organizations and/or whether they result in more profitable or strategically well-positioned firms whether or not they are more agile. It is of course difficult to tease out the discreet effects of project management practices (agile or otherwise) on large-scale organizational outcomes, largely because there are so many interconnected components of a large organization in operation at the same time and because there is generally no control group and variables become confounded (e.g., agile approaches to project management may be applied to a set of SWD projects with the traditional approach used in, for example, construction projects; thus we do not know whether effects are due to the practices or the match between practices and a particular environment. However, it stands to reason that if projects are delivered with on average better time and cost results, those benefits will accumulate organizationally to quicker response times, extra financial assets applicable to additional projects or other uses and the like.

The inverse perspective may also be useful for generating new knowledge. Will a more agile firm be more likely to create and/or benefit from agile approaches to project management? Perhaps agility at a firm level and project level evolve in a co-evolutionary pattern where successes at each level in the hierarchy prompt experimentation and extension at other levels.

Assuming that agile practices applied to projects do lead to organizational agility, this might come at the cost of diminishing optimal practices. To the extent that flexibility involves investing in options that may or may not be exercised, this investment has a cost, and options not exercised may pose a significant cumulative expense over time. It is also possible that an organization may benefit from the efficiencies of agile practices without increasing organizational agility. For example, agile practices may be used for updating an assembly line or supply chain process (in a more efficient manner) without affecting standard business process approaches to ongoing work.

It is worth remembering that the demand for flexibility was raised previously in the 1970s, and traditional project management was supposed to address the flexibility challenge, for example, by introducing flexible organizational structures such as matrix organizations (Galbraith, 1971). While perhaps

increasing flexibility, matrix organizations also introduced significant complexity particularly in terms of the boundary spanning aspects of individual worker roles. Olsson (2006) reviews multiple forms of project flexibility. This perspective was mainly driven by challenges of resource allocation and coordination between the routine and the project organization. One could therefore argue that the request for flexibility and suggested solutions are not new. Flexibility per se is not necessarily the differentiating characteristic between agile and traditional practices. Agile practices address a focused type of flexibility related to project goals and requirements. Thus far, flexibility in the requirements and the processes that are iterated until satisfying results are achieved has been enabled with a rigid resource structure, namely a core team. Further discussions are necessary to address the flexibility argument and better understand if trade-off compromises between organizational flexibility and task flexibility are unavoidable.

Alternatives to both agile and traditional approaches. It is also of interest to recall that traditional SWD practices arose from many sources in the early days of computing, when it was observed that ad hoc development of software resulted in very expensive maintenance costs due to the lack of maintainable code and documentation. Many attempts over the years to improve SWD by reducing the formality of traditional methods have followed. It is, however, with agile practices that an alternative approach has been widely adopted. Even with reported successes, software environments continue evolving, which may affect its fit with particular processes. Agile practices may have begun to be superseded by even newer approaches to software creation called “continuous development” and DevOps (Gruver & Mouser, 2015; Kim, Humble, Debois, & Willis, 2016). In this approach a larger development operation, often a platform, receives continuous updates. The traditional notion of a project with a clear start and end and well-defined objectives may dissolve into an ongoing stream of project-like activities that begin to look like a development assembly line emphasizing continual evolution of the platform rather than the creation of distinct and separate products. Some organizations are releasing software updates (where some of these updates represent new features, others enhancements, and still others operational efficiencies) up to multiple times a day. In such a setting, we must ask what we mean by a “project”—even though each addition or modification can be viewed as separately creating a unique product—and reconsider how traditional approaches and tools can be helpful, whether they have outlived usefulness, if they can be reshaped to better fit the task, and/or if new approaches and tools are needed (e.g., Raith, Richter, & Lindermeier, 2017).

This approach has the potential to redefine thinking about the relationship between one-time projects and on-going work flows (Harvey & Aubry, 2018). What would shifting to this sort of approach require and what immediate and long-term effects would it create? Metrics related to cost, time, and quality would likely shift from evaluation of the process for creating

individual products to operational measures of the platform—is it growing or losing users, is performance improving or declining, which new areas of growth should be followed, and are there extraneous features constraining other uses/activities? Team formation may highlight responsibility for a particular work domain including both ongoing operations and development of additional capabilities. Hiring criteria may require skill sets related to both maintenance and innovation. The quality of assignment of individuals to tasks, or their self-selection, may rise along with execution quality as a cause of performance outcomes. Skill building may shift toward anticipating future tasks. Periodic sessions such as stand up meetings may focus on tasks related to both continual and one-time actions.

It is relatively easy to imagine that such an integrated project-process approach, if successful in software development and maintenance, can be adjusted for use in manufacturing, supply chain, and perhaps new product development. It is less clear what shifts in formal agile practices as defined by current methodologies would be helpful to facilitate such a transfer.

Conclusion

The original thinking behind this special issue has always been to survey the current state of knowledge regarding agile practices as applied in the SWD environment and to examine in more depth shifting research attention to the application of agile practices outside of SWD. Additionally, in this article, we present a research framework derived from fairly standard research dimensions (level of analysis and functional area) and use this to suggest potential issues and questions for future investigation. We see an opportunity, especially for management scholars, as the extant literature seems heavily oriented toward individual and team/project levels with much room for extension in the organizational level of analysis. Similarly, we see an opportunity to shift from the behavioral and process perspectives toward increased focus on governance. Given the array of stakeholders and their varied interests relative to SWD and other task projects, we see a need for continued inquiry into the definition and measurement of project outcomes. We iterate that the categories in this model are not meant to be comprehensive nor mutually exclusive, but rather are intended to serve as a mechanism for surfacing issues and raising questions. Individual studies sometimes cross categorical borders particularly as amounts and types of behavior, process, and governance are likely to affect outcomes.

In this introduction we have also emphasized theory. By this we mean the development of testable general statements about relationships among project elements linked to rigorous empirical testing. This is not to say we don't see value in the engineering and proof of concept research, which can form an empirical base of understanding. We caution, however, that while a practice can be shown to potentially create value, it doesn't mean that it automatically will do so. It is important to understand how it creates value; how it fits with other

organizational practices; and how it may need inputs from its environment to be successful broadly, regularly, and in the complex world of practice. Along with Dingsøyr et al. (2012) we emphasize the potential added value from a theory-based empirically validated set of research practices.

We have attempted to present a wide variety of possibilities for research questions and topics that promise to extend our knowledge of the application of agile practices across task domains. We also observe particular trends emerging toward further consideration of agile practices related to portfolios and programs, large projects, individual practices in much more depth, and hybrids as well as other alternatives to either agile or traditional practices.

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Notes

1. Gartner Inc. is a consulting firm that tracks the diffusion of a wide range of information technology products, services, and concepts. They track these along an observed recurrent pathway by which initial excitement about a product grows until reaching a peak, then descends rapidly only to be replaced by slow and steady growth where it is appropriate, rather than as a perceived panacea. For an example of a particular hype cycle analysis, see: https://www.bing.com/images/search?view=detailV2&ccid=5oE8JZFT&id=AF3036BD626445674CB08CD57EE21C31178C3062&thid=OIP.5oE8JZFTKFB0Gr_0io-e5wHaGR&mediaurl=http%3a%2f%2fblogs.gartner.com%2fsmarterwithgartner%2ffiles%2f2017%2f08%2fEmerging-Technology-Hype-Cycle-for-2017_Infographic_R6A.jpg&exph=1269&expw=1500&q=gartner+hype+cycle+2018&simid=608039093487208347&selectedIndex=0&qpvrt=gartner+hype+cycle+2018&ajaxhist=0
2. We use the term “agile practices” consistently throughout the article rather than an equivalent term “agile methods” in order to distinguish the practice of agile approaches to project management from the ambiguity of methods applied to research procedures as opposed to project actions.
3. Based on search of ABI Informs 3 March 2018.

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